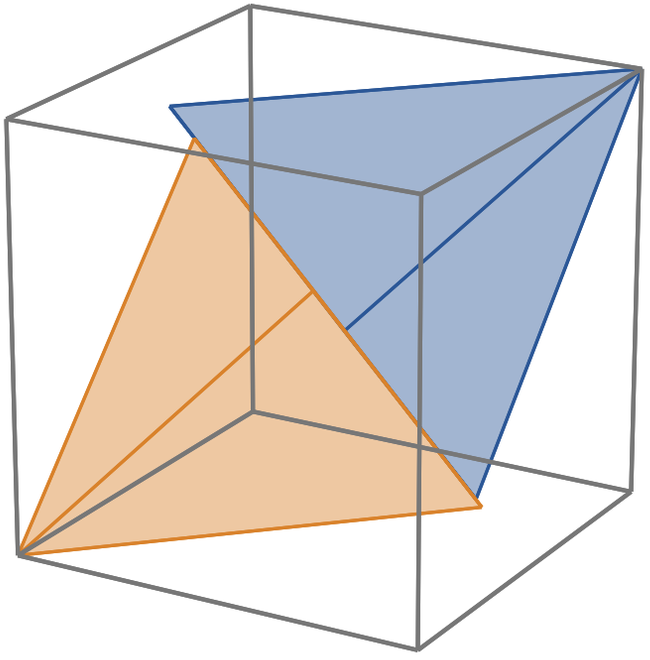
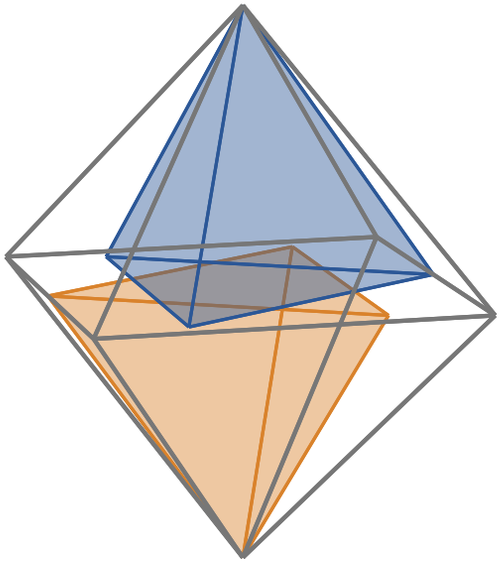
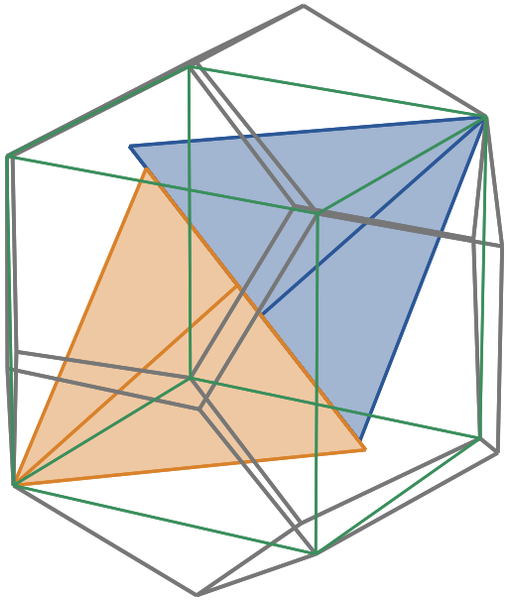
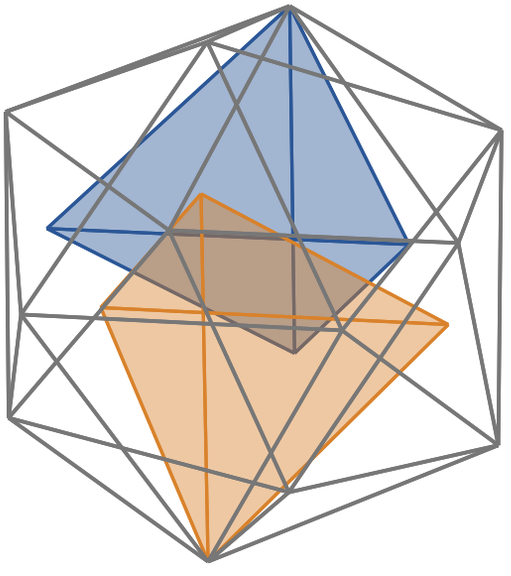
cube, side $2\sqrt{2}/3 \approx 0.943$



octahedron, edge $2/\sqrt{3} \approx 1.155$



dodecahedron, edge $\sqrt{2}(\sqrt{5}-1)/3 \approx 0.583$



icosahedron, edge $2\sqrt{(5-\sqrt{5})/15} \approx 0.859$